

Benchmark Report

Benchmarking of NQCH’s quantum computer

April 16, 2026

1. Report of Changes

Platform: sinq20

Calibration-id: ee177a7b10578056b733df2a2d86a1a3fda5f287

Calibration date: 2026-04-16 01:23:19

Calibration note: chore(sinq20): re-initial coarse calibration

Experiment-id: 20260416092939

Experiment date: 2026-04-16 09:29:39

Experiment note: temporary note!!!

Platform: sinq20

Calibration-id: 4dc4082f38a53222b3956c22202d32a520d4bc78

Calibration date: 2026-01-15 02:03:03

Calibration note: chore(sinq20): 2q gates cal 0-1, 0-3, 2-3, 3-4, 1-4, 4-5, 4-9, 3-8, 8-9

Experiment-id: 20260118161538

4. Statistics

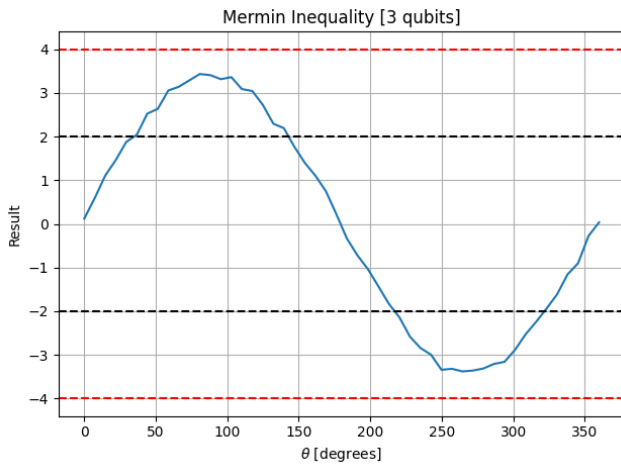
	Average	Median	Min	Max
T1 (ns)	23.3	23.4	16.2	35.9
T2 (ns)	12.6	11.7	1.08	32.6
Fidelity	0.994	0.997	0.981	0.998
RO Fidelity	0.986	0.987	0.979	0.991
2q RB Fidelity	-	-	-	-
2q CZ Fidelity	-	-	-	-

	Average	Median	Min	Max
T1 (ns)	1.28e+04	1.23e+04	646	3.65e+04
T2 (ns)	2.36e+25	3.68e+03	125	9.43e+26
Fidelity	0.99	0.997	0.887	0.999
RO Fidelity	0.794	0.777	0.777	0.927
2q RB Fidelity	0.965	0.972	0.923	0.981

8. Mermin

Mermin's algorithm for 3 qubits.

Runtime	Qubits	Max
603.71 sec	11, 5, 6	3.432



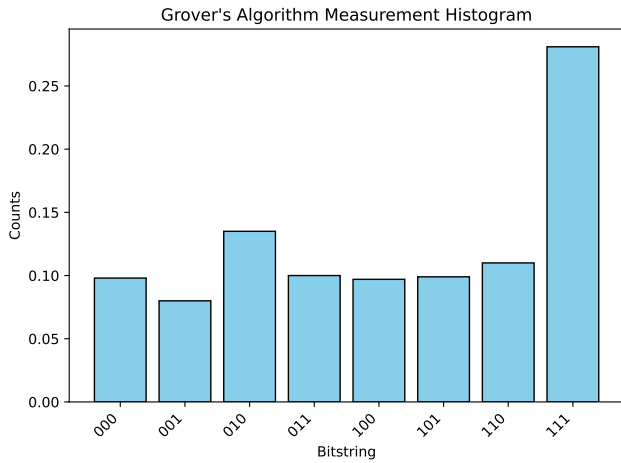
Runtime	Qubits	Max
1.02 sec	0, 2, 3	-3.316



10. Grover - 3 qubits

Grover's algorithm for 3 qubits executed on sinq20 backend with 1000 shots per circuit. We measure the success rate of finding the target state '111' for each pair of qubits in [[11, 6], [5, 6]].

Runtime	Qubits	Max prob
2.44 sec	11, 5, 6	0.281



Runtime	Qubits	Max prob
2.43 sec	0, 2, 3	0.385